METRIC

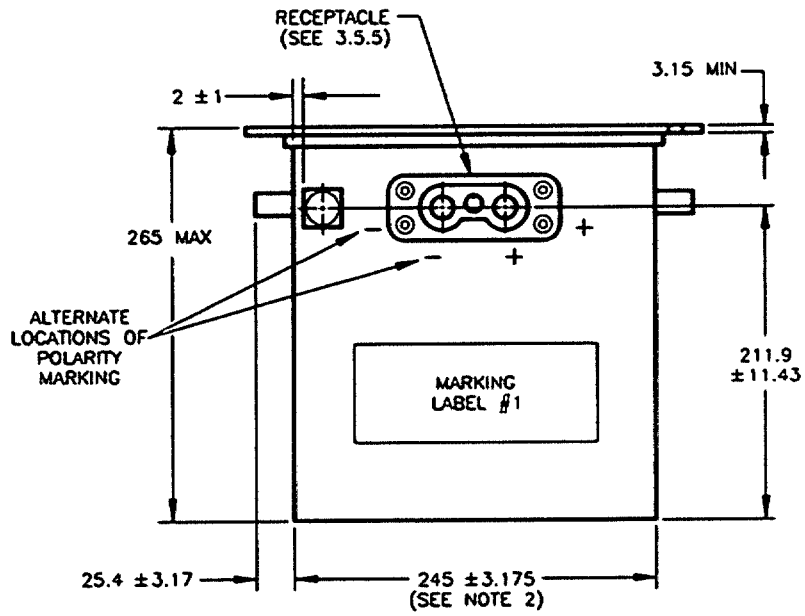
MIL-B-8565/12
17 May 1993

MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1, MAINTENANCE-FREE, 24-VOLT, 35-AMPERE-HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:
MIL-B-8565



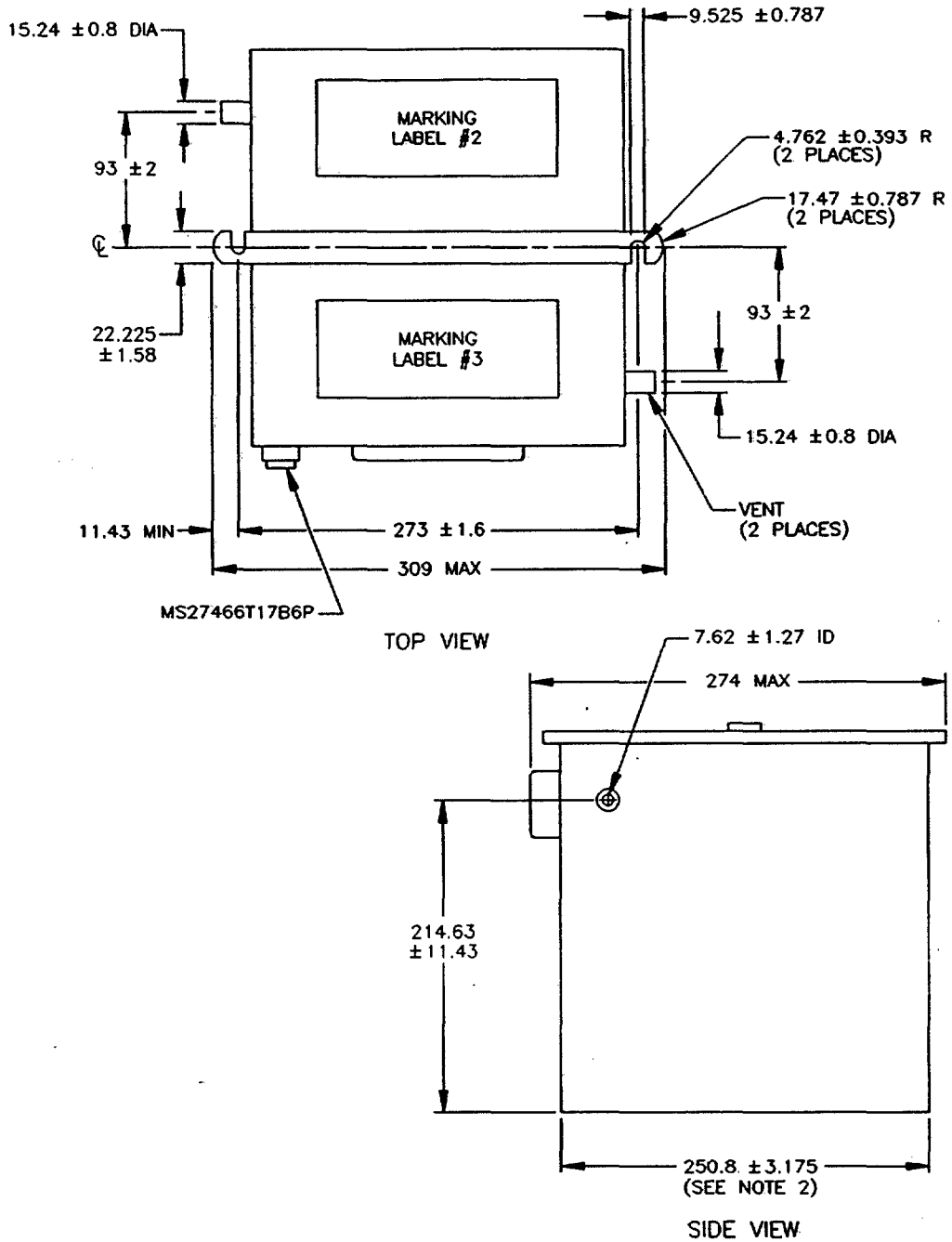
FRONT VIEW

NOTES:
1. All dimensions are in millimeters.
2. Allow additional 12mm width for cover.
3. Abbreviations used in this figure are maximum (MAX), minimum (MIN) and part number (PN).

FIGURE 1. PN D8565/12-1 dimensions.

AMSC N/A FSC 6140
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.



NOTES:

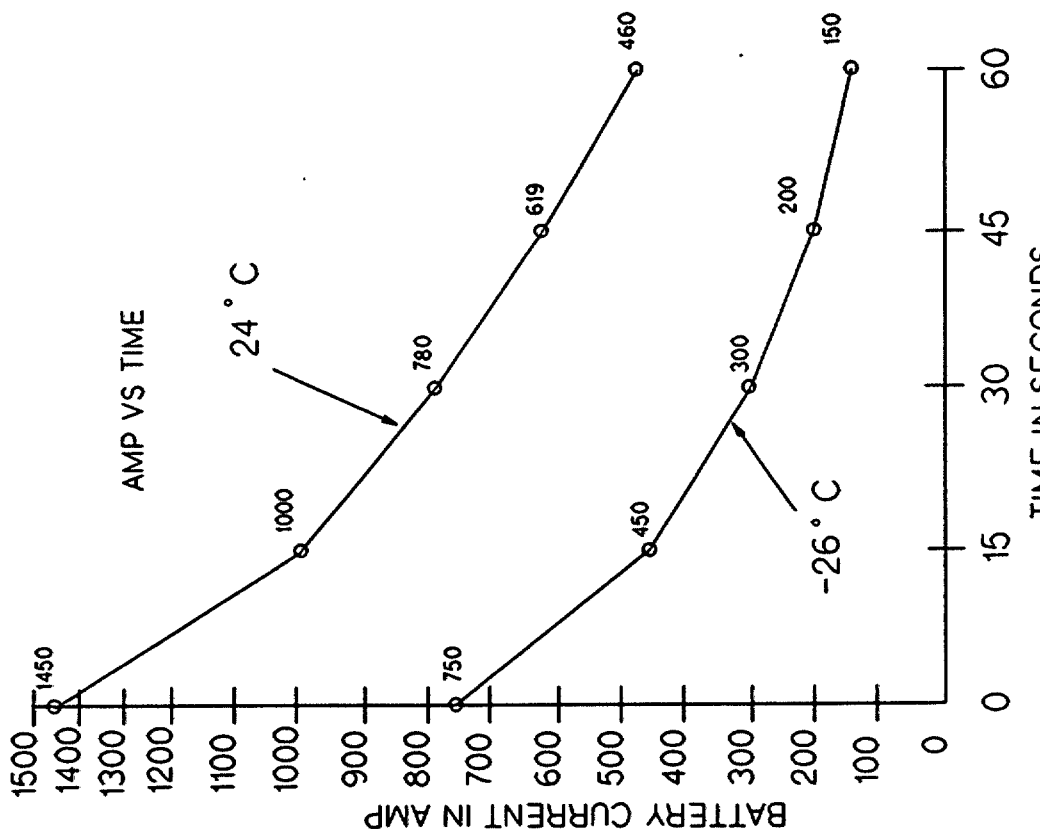
1. All dimensions are in millimeters.
2. Allow additional 12mm length for cover.
3. Abbreviations used in this figure are diameter (DIA), inside diameter (ID), maximum (MAX), minimum (MIN), radius (R) and part number (PN).

FIGURE 1. PN D8565/12-1 dimensions-Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.

DISCHARGE CURRENT VS TIME		
TIME IN SECONDS	CURRENT IN AMP	
	AT 24°C	AT -26°C
0+	1450	750
15	1000	450
30	780	300
45	619	200
60	460	150

AVERAGE DISCHARGE CURRENT DURING 14-V DISCHARGE SHALL BE NOT LESS THAN THE FOLLOWING AMP VALUES	
24°C	838.0 AMP
-26°C	350.0 AMP

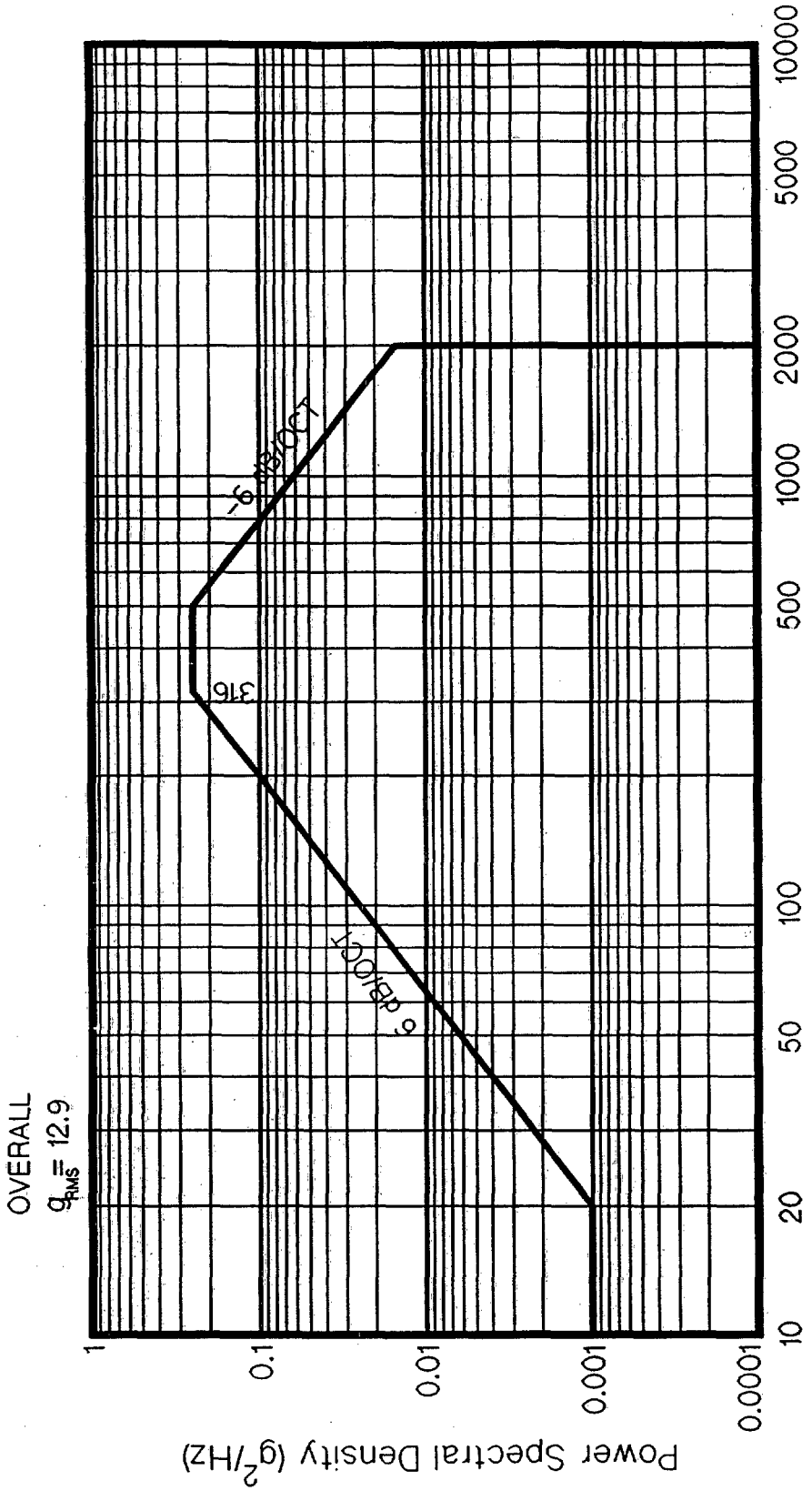


NOTE:

1. Abbreviations used in this figure are amperes (AMP) and versus (VS).

FIGURE 2. Constant 14-V discharge curves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.



NOTE:

1. Abbreviations used in this figure are acceleration of gravity (g), decibel (dB), root mean square (RMS), hertz (Hz) and octave (OCT).

FIGURE 3. Random vibration test levels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. The dimensions of the battery shall be as shown on figure 1.

2. The part number of the battery shall be D8565/12-1.

3. The rated capacity of the battery shall be 35 ampere-hours, based on its ability to conform to the life test final discharge. The life test final discharge is a 1-hour discharge at 24°C ambient temperature with a 35-ampere load (1C-rate) to a cutoff voltage of 18 volts.

4. The weight of the battery shall be not greater than 40.8 kg.

5. For installation in the aircraft, the battery shall be supported as shown in the conventional mounted example on figure 1 of MIL-B-8565. Battery life in the aircraft shall be 4 years.

6. The battery shall be constructed such that performance is not affected by sand and dust. Battery materials shall not support fungus growth.

7. The battery shall not cause an explosion when operated in an explosive atmosphere.

8. Table I lists the quality assurance tests.

9. Delete all requirements for cells and delete all requirements for threaded-post connected batteries.

10. The battery shall conform to MIL-B-8565 for type I, medium rate, except as modified by the following.

3.5.5 Receptacles. Add the following: "The receptacles shall be located as shown on figure 1 of the applicable specification sheet. The receptacle for the battery's positive and negative connections shall be MS3509, style 1 or equivalent. The auxiliary receptacle, MS27466T17B6P, shall be connected as follows:

a. Pin A shall be the 115 volts (V) alternating current (AC) (input heater excitation).

b. Pin B shall be the AC ground (heater return).

c. Pin C shall be the temperature sensor.

d. Pin D shall be the temperature sensor.

e. Pin E shall be the heater test point.

f. Pin F shall be the case ground."

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.

3.5.12 Reliability. Delete "1500 hours" and substitute "3000 hours".

Add as a new requirement:

"3.5.13 Temperature environment. The operating temperature environment shall be -40°C to 71°C, while the non-operating environment shall be -54°C to 85°C."

3.6.3.2.1 Polarity marking. Add the following: "The marking for label 3 shown on figure 1 of the applicable specification sheet shall show the following with a lettering height not less than 0.15 inch:

PIN A 115 VAC HEATER EXCITATION

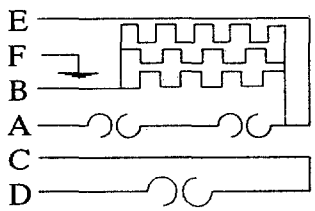
PIN B HEATER RETURN

PIN C TEMP. SENSOR

PIN D TEMP. SENSOR

PIN E HEATER TEST POINT

PIN F CASE GROUND



HEATER WIRING SCHEMATIC

Centered at the top of the label in bold letters with a height not less than 0.25 inch shall be the words "CAUTION HOT SURFACE"."

3.6.3.2.3 Battery caution marking (type 1 battery only). Delete the following: "except for "WARRANTED ITEM" line which shall be 0.375 inch minimum."

3.6.8 Capacity and electrical performance. Add the following: "During the 1-minute, 14-volt, constant voltage discharge, the battery shall meet or exceed the current values shown on figure 2 of the applicable specification sheet throughout the first 15 seconds of discharge. Furthermore, through the duration of the discharge, the battery shall maintain an average current not less than the average value shown on figure 2 of the applicable specification sheet. Inability of the battery to deliver the specified currents during the discharge shall constitute a test failure."

TABLE II. Capacity performance requirements (type 1 battery). Add the following test requirements as indicated:

Requirement Number	Discharge Cutoff Volts & Temp.	Rate of Discharge Factor	Minimum Time to Cutoff Voltage
"(6)	18.0 volts at -40°C	15 amperes	58 minutes
(7) See 2/	18.0 volts at -40°C	36 & 100 amperes	40 & 15 seconds

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.

2/ Prior to conducting this capacity test, the battery shall be fully charged and then stand for 14 days on open circuit at 32°C. The battery shall be subjected to a two-step discharge as follows: discharge at 36 amperes for 40 seconds and then immediately increase current to 100 amperes. The battery shall maintain a terminal voltage of 18 volts or greater for 40 seconds at 36 amperes and for a minimum of 15 seconds at 100 amperes."

3.6.10 Life. Delete "2.5 years" and substitute "4 years".

Add as a new requirement:

"3.6.23 Cold and hot temperature charging.

"3.6.23.1 Half-hour charging. The battery, when tested as specified in 4.6.20.2.1, having 17.5 ampere-hours at temperatures in the range of -18°C to 71°C, shall be capable of delivering not less than 26.25 ampere-hours capacity after a 28.25 ± 0.5 volt charging source has been applied to the battery for 30 minutes.

"3.6.23.2 One-hour charging. When tested as specified in 4.6.20.2.2, the battery, when discharged at temperatures in the range of -40°C to -18°C at a current of 35 amperes to 18 volts, shall be capable of delivering not less than 26.25 ampere-hours capacity after a 28.25 ± 0.5 volt charging source has been applied to the battery for 1 hour. The battery shall not be damaged when a 28.25 ± 0.5 volt charging source is applied for 1 hour to the battery at -40°C.

"3.6.24 Heaters. Heaters may be incorporated into the battery to meet the cold temperature charging requirements. The heaters shall utilize 115 VAC power in accordance with MIL-STD-704. Heater power shall be accessible at auxiliary connector pins A and B. Heater current shall not exceed 5 amperes at 115 VAC. Heater power shall not be provided during any discharge performed in accordance with this specification. Heater power shall be provided during all constant-potential charges.

"3.6.24.1 Heater element. The heater element shall be fabricated and installed in the battery assembly using insulated heaters and wire assemblies so as to prevent the possibility of sparking.

"3.6.24.2 Temperature limiting device. The battery shall contain temperature limiting devices as needed to prevent overheating when the heater circuit is energized. Under any condition of operation specified herein at an ambient temperature of 25 ± 5°C, the battery case temperature shall not exceed 75°C due to heater and battery operation. The statement "CAUTION HOT SURFACE" shall be shown on marking label 3. The heater control devices shall be compatible with the battery heater current requirement. If thermal switches are used, they shall be in accordance with MIL-S-24236, type I, class 4 or higher. The switches shall be designed and installed to prevent ignition of an explosive mixture of hydrogen and oxygen. All components of the heater circuit shall be located inside the battery case. In the event that the heater control fails, the battery shall not reach a temperature which will damage the battery in any way.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.

"3.6.25 Safe charge limit switch. The battery shall contain one thermal switch to provide contact closure for the battery charger through the auxiliary connector when the battery temperature exceeds the safe charging limit. The thermal switch shall close at 79 ± 5°C and open at 68 ± 5°C. The switch shall be in accordance with MIL-S-24236, type I, class 4 or higher and shall be designed and installed to prevent ignition of an explosive mixture of hydrogen and oxygen. The switch shall have a dry circuit contact rating of less than 5 milliamperes.

"3.6.26 Reference temperature. The manufacturer shall determine a location which is representative of the average cell temperature for the temperature sensing switches specified in 3.6.24.2.

"3.6.27 Short circuit current. The battery, including all of its current carrying parts, shall withstand a 1500-ampere discharge for not less than 30 seconds without creating a hazard such as an explosion or fire when tested as specified in 4.6.20.3."

4.5.1.3 Constant-potential charge. Delete "2.0 ± 0.1 hours" and substitute "4.0 ± 0.1 hours". Add the following: "Heater power will be provided during all constant-potential charges."

4.5.1.5 Discharge. Add the following: "If the battery contains a heater circuit, no power will be applied to the heater circuit during any discharge."

4.6.5 Dielectric strength. Delete and add the following: "The insulation resistance between the current carrying parts of the battery, heater circuit, safe charge limiting circuit, and battery container shall be not less than one megohm when 500 ± 10 VDC is applied to each pin or terminal of each circuit and between each pin or terminal of all other circuits and exposed metal of the container. The battery shall meet the specified requirement of 3.6.4."

4.6.13.1 Cycling test. Delete "2-hour charge" and substitute "4-hour charge".

4.6.20 Special tests. Add the following:

"4.6.20.1 Capacity discharge at -40°C. The battery shall be charged as specified in 4.5.1.3 and stored under the specified temperatures and storage conditions (see 4.5.1.2). The battery shall then be discharged at the rate specified in accordance with requirement (6) of table II to an end voltage of 18 volts. The battery shall meet the requirement of 3.6.8. Repeat the above procedures except the battery shall undergo a 2-step discharge at the specified rate in accordance with requirement (7) of table II.

"4.6.20.2 Cold and hot temperature charge tests.

"4.6.20.2.1 Half-hour charging. Battery sample 2 shall be tested as follows to verify conformance to paragraph 3.6.23.1.

a. Charge the battery at 28.25 ± 0.5 volts for 4 hours at 25°C.

b. Stabilize the battery at -18°C, then discharge at 35 amperes to 18 volts. The battery

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.

shall provide at least 17.5 ampere-hours capacity.

c. Stabilize the battery at 25°C, then charge at 28.25 ± 0.5 volts for 4 hours.

d. Stabilize the battery at -18°C, then discharge at 35 amperes to remove all but 17.5 ampere-hours, as determined above by 4.6.20.2.1.b.

e. Rest the battery and stabilize at -18°C, then charge at 28.25 ± 0.5 volts for 30 minutes.

f. Immediately following charge, remove battery the from the test chamber and discharge at 35 amperes to 18 volts. The battery must deliver not less than 26.25 ampere-hours.

g. Repeat steps a. through f. except at a test temperature of 71°C.

"4.6.20.2.2 One-hour charging. Battery sample 3 shall be tested as follows to verify conformance to paragraph 3.6.23.2.

a. Charge the battery at 28.25 ± 0.5 volts for 4 hours at 25°C.

b. Stabilize the battery at -40°C, then discharge at 35 amperes to 18 volts.

c. Rest the battery and stabilize at -40°C, then apply a 28.25 ± 0.5 volt charge for 1 hour.

d. Immediately following charge, remove the battery from test chamber and discharge at 35 amperes to 18 volts. The battery shall deliver not less than 26.25 ampere-hours.

"4.6.20.3 Short circuit current test. A fully charged battery, within the confines of a temperature chamber, shall be subjected to a constant current discharge rate of 1500 amperes for a period of 30 seconds. At the conclusion of testing, the battery shall be examined for conformance to the requirements of paragraph 3.6.27.

4.6.20.4 Temperature limiting device test. To determine conformance to the requirements of paragraphs 3.6.25, the following test shall be performed. The battery cover shall be removed and the battery instrumented by the addition of thermocouples to the battery adjacent to the hot battery thermoswitch. Operation of the hot battery thermoswitch shall be monitored between pins C and D of the auxiliary connector using an ohmmeter. Operating temperatures shall be determined by monitoring the added thermocouples. The battery cover shall be attached during the test.

a. Place the battery, with the heater circuit energized, into a test chamber stabilized at 65 ± 2°C.

b. Increase the chamber temperature to 85 ± 2°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.

c. Monitor the hot battery thermoswitch to determine the temperature at which the first switch closes.

d. After the hot battery thermoswitch has closed, immediately reduce the test chamber temperature to 50 ± 2°C.

e. Monitor the hot battery thermoswitch to determine at what temperature the first switch re-opens.

4.6.23 Vibration. Delete and add the following: "Endurance and performance random test. The battery assembly, including connectors and cable, shall be mounted to the test apparatus, per 4.5.5 and figure 1 of the general specification, and subjected to the random vibration test procedure for equipment installed in jet aircraft, Category 5 Procedure I, Method 514.4 of MIL-STD-810. The battery shall be vibrated along each orthogonal axis in accordance with the Random Vibration Test levels specified in figure 3 of the applicable specification sheet for a period of 6 hours per axis. The battery heater shall be energized and the battery charged at 28.25 ± 0.5 volts during vibration and shall be discharged at 6.5C for 1 minute before beginning each axis of vibration. The cables connected to the battery shall consist of four size AN-6 cables (two connected to the positive terminal and two connected to the negative terminal) and shall be clamped to the test apparatus at a cable length of 46 ± 3 centimeters from the battery."

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.

TABLE I. Inspection of batteries and order of test.

Order of test	Description of inspection	Requirements paragraph	Inspection method paragraph	Qualification inspection			Quality conformance inspection		
				Qualification sample number			Group A tests	Group B tests	
				1	2	3		1	2
1	Visual & mechanical examination	3.5, 3.6.1 & 3.6.3	4.6.3	X	X	X	X	X	X
2	Dimensions & weight	3.6.5	4.6.4	X	X	X	X	X	
3	Dielectric strength	3.6.4	4.6.5	X	X	X	X	X	X
4	Strength of vent tubes	3.6.7	4.6.7	X	X	X	X	X	X
5	Color & marking	3.6.3	4.6.8	X	X	X	X	X	X
6	Conditioning charge	4.5.1.4	4.6.9	X	X	X		X	X
7	Capacity discharge	3.6.8	4.6.10	X	X	X	X	X	X
8	Constant voltage discharge	3.6.8	4.6.11	X	X		X	X	
9	Strength of receptacle	3.6.9	4.6.12			X			X
10	Capacity discharge at -40°C	3.6.8	4.6.20.1		X				X
11	Cold & hot temperature charge test	3.6.23	4.6.20.2		X	X			X
12	Life cycling	3.6.10	4.6.13	X				X	
13	Capacity discharge @ 49°C	3.6.8	4.6.14		X				
14	Discharge while inverted	3.5.8 & 3.6.8	4.6.16		X				X
15	Altitude	3.6.11 & 3.6.12	4.6.17		X				X
16	Mechanical shock	3.6.11 & 3.6.12	4.6.18	X					
17	Temperature shock	3.6.11 & 3.6.12	4.6.19	X				X	
18	Temperature rise & float	3.6.13	4.6.21		X				X
19	Battery gas emission test	3.6.14	4.6.22		X				
20	Vibration	3.6.16	4.6.23		X				
21	Humidity	3.6.17	4.6.24		X				X
22	Temperature limiting devices	3.6.24.2 & 3.6.25	4.6.20.4		X				
23	Salt fog	3.6.18	4.6.25	X				X	
24	Ground storage	3.6.15.1	4.6.26			X			
25	Short circuit current	3.6.27	4.6.20.3			X			
26	Physical integrity at high temperature	3.6.12	4.6.27	X					
27	Final examination	3.6.4	4.6.30	X	X	X		X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.

Custodian:
Air Force - 99
Navy - AS

Preparing activity:
Navy - AS

(Project 6140-0753)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.